

Argument ellipsis in Russian verb-stranding: an acceptability study of semantic constraints

Gleb Tumin

gleb.tumin@gmail.com

March 22, 2026

Abstract

Russian verb-stranding constructions, in which a finite verb appears without its internal argument, have been analyzed as instances of either Verb-stranding VP Ellipsis (VVPE) or Argument Ellipsis (AE). Recent work has established that AE is restricted to arguments of semantic type *e*, providing a diagnostic: if verb-stranding shows degraded acceptability with non-*e*-type arguments, AE rather than VVPE is the operative mechanism. This study extends prior experimental evidence for AE in Russian (Kasenov & Rudnev, to appear) by decomposing the non-*e*-type category into four subtypes: NAMING COMPLEMENT, PREDICATIVE NOMINAL, PREDICATIVE ADJECTIVAL, and ARGUMENTAL ADVERBIAL. In a pre-registered acceptability judgment experiment ($N = 260$), the *e*-type baseline (DP OBJECT) was rated significantly higher than three of the four non-*e*-type conditions, with medium-to-large effect sizes ($d = 0.57\text{--}0.83$). The contrast between DP OBJECT and PREDICATIVE NOMINAL did not reach significance ($d = 0.25$, $p = .133$); however, the post-hoc analysis indicates that the study was underpowered to detect effects of this magnitude, leaving open whether predicative nominals pattern with *e*-type or non-*e*-type arguments. The results of the study provide further support for AE as the primary analysis for Russian verb-stranding and raise the question of whether predicative nominals admit an individual-denoting reading that renders them compatible with argument ellipsis.

1 Introduction

Sentences such as (1), in which a finite verb is used without an overt internal argument, are often referred to as the verb-stranding constructions (Goldberg, 2005):

- (1) Andrej pomyl mašynu, i Boris tože pomyl.
Andrej.NOM wash.PST.SG.M car.ACC.SG and Boris.NOM too wash.PST.SG.M
'Andrej washed the car, and Boris did too.'

The theoretical question raised by (1) is what licenses the absence of the internal argument. In Russian syntax, two competing analyses differ in the proposed mechanism licensing the absent argument. Under Verb-stranding VP ellipsis (VVPE) (Gribanova, 2013b,a), the lexical verb *V* raises to *v*, forming the verbal complex, which then raises above *vP* (to Asp/T); *vP* is subsequently deleted, stranding the verbal complex. Under Argument Ellipsis (AE), an alternative approach (Oku, 1998; Saito, 2007; Takahashi, 2006, 2008) (see also Bošković (2018); Landau (2020) among others; and Bailyn (2017); Bailyn & Bondarenko (2021) for Russian), individual arguments are absent from the syntactic derivation.

Although both theories based on either (a) deletion of the absent material at Phonological Form (PF deletion) or (b) interpreting the empty argument position at Logical Form via copying of a contextually available antecedent (LF copying) do not predict constraints on the semantic type of the elided argument, it has been shown (Landau (2018) for Hebrew, Landau (2020, 2023b) in general) that AE can only target *e*-type arguments. As no such effect has been observed for VVPE (as it is primarily analyzed as (a)), the E-TYPE diagnostic follows naturally: an E-TYPE effect – significantly lower acceptability for verb-stranding constructions with non-*e*-type absent arguments – would demonstrate that

AE, not VVPE, is taking place in a verb-stranding construction, establishing it as a superior analysis for Russian verb-stranding. The absence of E-TYPE effect, on the other hand, would demonstrate that VVPE is likely to be the primary analysis in this case, though would not constitute a direct evidence for it, as it is also compatible with AE under a weaker version of the semantic type constraint.

Kasenov & Rudnev (2024) provide experimental evidence in favor of the AE approach to Russian verb-stranding, reporting a significant E-TYPE effect (Experiment 2: coefficient = .510, SE = .088, $p < .001$) in an acceptability judgment study. However, their design does not distinguish between different types of non-*e*-type arguments. The present study extends the line of inquiry by decomposing the non-*e*-type category arguments into four distinct subtypes: NAMING COMPLEMENT, PREDICATIVE NOMINAL, PREDICATIVE ADJECTIVAL, and ARGUMENTAL ADVERBIAL, based on types mentioned in Landau (2023a) as those constraining AE, and comparing them with an *e*-type argument: DP OBJECT. This approach makes it possible to control whether the E-TYPE effect, if found, is pure for all the *e*-type vs. non-*e*-type pairs or is subject to the specific semantic type of the non-*e*-type argument.

The hypotheses and a research question structuring the analysis are as follows.

Confirmatory (Holm–Bonferroni corrected; family-wise $\alpha = .05$):

H1a: Acceptability ratings for DP OBJECT will be significantly higher than acceptability ratings for NAMING COMPLEMENT.

H1b: Acceptability ratings for DP OBJECT will be significantly higher than acceptability ratings for PREDICATIVE NOMINAL.

H1c: Acceptability ratings for DP OBJECT will be significantly higher than acceptability ratings for PREDICATIVE ADJECTIVAL.

H1d: Acceptability ratings for DP OBJECT will be significantly higher than acceptability ratings for ARGUMENTAL ADVERBIAL.

Exploratory (Holm–Bonferroni corrected; family-wise $\alpha = .05$):

RQ1: What is the acceptability profile across the non-*e*-type argument types?

The results of the experiment demonstrate that the semantic type of the absent argument affects the acceptability ratings, thus providing evidence for AE as the primary analysis for Russian verb-stranding.

Section 2 describes the methods of the experiment. Section 3 describes the results. Section 4 discusses the observed results. Section 5 concludes.

2 Methods

2.1 Participants

301 adult native speakers of Russian were recruited via Yandex.Tasks crowdsourcing service (with native-language screening), and directed to the experiment via a link to PCIBex Farm (Zehr & Schwarz, 2018). All participants were self-reported monolingual native speakers of Russian with no reported neurological or speech disorders. Participation via Yandex.Tasks was compensated in accordance with the platform’s standard rate. The experiment took approximately 10 minutes to complete.

2.2 Design

The experiment employed a single-factor within-participants design with five levels of the factor CONDITION: DP OBJECT, NAMING COMPLEMENT, PREDICATIVE NOMINAL, PREDICATIVE ADJECTIVAL, and ARGUMENTAL ADVERBIAL. Experimental items were distributed across participants using a Latin square counterbalancing procedure. The item pool consisted of 8 items per condition (40 target items in total), divided into 4 lists of 10 target items each (2 per condition per list). Each participant was thus exposed to all five conditions, encountering 2 target items per condition. Target items were presented alongside filler items at a 1:2 ratio (10 target items and 20 filler (10 *lower* anchors and 10 *upper* anchors) items per participant).

2.3 Materials

2.3.1 Target items

Target items instantiated Russian matrix verb-stranding constructions, in which the second conjunct contains the verb without its argument, licensed by an identical argument in the antecedent conjunct. The additive particle *tože* 'too/also' in the second conjunct establishes an additive discourse context, in which focus falls on the subject rather than on the polarity of the proposition, making a polarity ellipsis parse (TP-ellipsis; [Gribanova \(2017\)](#)) pragmatically unlikely.

The five conditions differed in the semantic type of the absent argument. The DP OBJECT condition (2) served as the *e*-type baseline, featuring transitive verbs with ordinary individual-denoting DP objects (i.e., referential direct objects realized as full noun phrases). The remaining four conditions instantiated non-*e*-type arguments of different morphosyntactic kinds: NAMING COMPLEMENTS (3), PREDICATIVE NOMINALS (4), PREDICATIVE ADJECTIVALS (5), and ARGUMENTAL ADVERBIALS (6).

(2) DP OBJECT (*e*-type)

Včera v buxgalterii glavbux proveril otčet, i ego
yesterday in accounting chief.accountant.NOM check.PST.SG.M report.ACC.SG and his.POSS
zamestitel' tože proveril.
deputy.NOM.SG.M too check.PST.SG.M

'Yesterday in the accounting department, the chief accountant checked the report, and his deputy did too.'

(3) NAMING COMPLEMENT (non-*e*-type)

Miša nazval kota Piratom, i Kolja tože
Misha.NOM name.PST.SG.M cat.ACC.SG.M Pirate.INST.SG.M and Kolya.NOM too
nazval.
name.PST.SG.M

'Misha named the cat Pirate, and Kolya did too.'

(4) PREDICATIVE NOMINAL (non-*e*-type)

Posle stažirovki Maša stala koordinatorom, i Ira
after internship.GEN.SG Masha.NOM become.PST.SG.F coordinator.INST.SG.M and Ira.NOM
tože stala.
too become.PST.SG.F

'After the internship, Masha became a coordinator, and Ira did too.'

(5) PREDICATIVE ADJECTIVAL (non-*e*-type)

Posle otpuska Il'ja stal spokojnym, i Denis tože
after vacation.GEN.SG Ilja.NOM become.PST.SG.M calm.INST.SG.M and Denis.NOM too
stal.
become.PST.SG.M

'After the vacation, Ilja became calm, and Denis did too.'

(6) ARGUMENTAL ADVERBIAL (non-*e*-type)

Na sobesedovanii Il'ja vël sebja sderžanno, i Dima tože
at interview.LOC.SG Ilja.NOM behave.PST.SG.M REFL restrainedly and Dima.NOM too
vël.
behave.PST.SG.M

'At the job interview, Ilja behaved restrainedly, and Dima did too.'

2.3.2 Filler items

Filler items served two purposes: to provide calibration anchors for participant exclusion and to prevent strategic response patterns. Fillers were divided into *upper* anchor and *lower* anchor items. *Upper* anchor items were well-formed sentences expected to receive high acceptability ratings; *lower* anchor items were sentences containing morphosyntactic errors expected to receive low ratings.

An example of an *upper* anchor filler is given in (7), and an example of a *lower* anchor filler is given in (8).

(7) UPPER ANCHOR (grammatical)

Kogda my pošli v gosti, Boris prinjës cvety, a
 when we.NOM go.PST.PL to visit.ACC.PL Boris.NOM bring.PST.SG.M flowers.ACC.PL and
 Kirill prinjës konfety.
 Kirill.NOM bring.PST.SG.M candy.ACC.PL

‘When we went visiting, Boris brought flowers, and Kirill brought candy.’

(8) LOWER ANCHOR (ungrammatical: case mismatch)

* Žora nazval kotěnkóm Zajka, i Maks tože
 Zhora.NOM name.PST.SG.M kitten.INST.SG.M Bunny.NOM.SG and Maks.NOM too
 nazval kotěnkóm.
 name.PST.SG.M kitten.INST.SG.M

Intended: ‘Zhora named the kitten Bunny, and Maks did too.’

2.4 Procedure

Participants rated each sentence on a 7-point Likert scale (1 = completely unnatural, 7 = completely natural) via PCIBex Farm (Zehr & Schwarz, 2018). A training block of two items (one *upper* anchor and one *lower* anchor) with written feedback preceded the main task to familiarize participants with the scale and task format. Stimulus order was pseudorandomized by the standard PennController `rshuffle()` function.

3 Results

3.1 Participants and data quality

A total of 301 participants completed the experiment. Participants were excluded on two pre-registered criteria applied to filler trials. By *criterion (A)*, participants whose responses to reliable anchor fillers were inconsistent with the expected direction on more than 20% of trials (rating ≤ 3 for *upper* anchors; rating ≥ 5 for *lower* anchors) were excluded ($n = 39$). Prior to applying this criterion, *upper* and *lower* anchors reliability was assessed at the item level: only items with mean rating ≥ 5 for *upper* anchor or ≤ 3 for *lower* anchor were used. Based on the rule, all the anchors were considered reliable: *upper* anchors obtained a mean rating of $M = 5.39$ – 5.64 and *lower* anchors of $M = 2.03$ – 2.52 , confirming that all anchor items fell within the expected range. By *criterion (B)*, participants whose overall rating standard deviation fell below 1.0, indicative of a flat-lining response pattern, were excluded ($n = 11$; overlap with the first criterion: $n = 9$). In total, 41 participants were excluded, leaving 260 participants for analysis ($N_{\text{obs}} = 2600$ target trials, 520 per condition).

3.2 Descriptive statistics

Raw and z-score-normalized mean acceptability ratings for each condition are provided in Table 1 and illustrated in Figure 1 and Figure 2. The DP OBJECT condition received the highest ratings (mean_{raw} = 4.61, $\bar{z} = 0.29$), followed by PREDICATIVE NOMINAL (4.24, $\bar{z} = 0.14$). The remaining

Table 1: Descriptive statistics by condition ($N = 520$ per condition).

Condition	n	M_{raw}	SD	SE	Median	$\bar{z}$
DP object	520	4.61	1.68	0.07	5	0.29
Predicative nominal	520	4.24	1.63	0.07	4	0.14
Predicative adjectival	520	3.84	1.72	0.08	4	-0.03
Naming complement	520	3.61	1.74	0.08	3	-0.15
Argumental adverbial	520	3.52	1.76	0.08	3	-0.17

Figure 1: Mean acceptability ratings ($\pm 95\%$ CI) by condition (raw ratings).

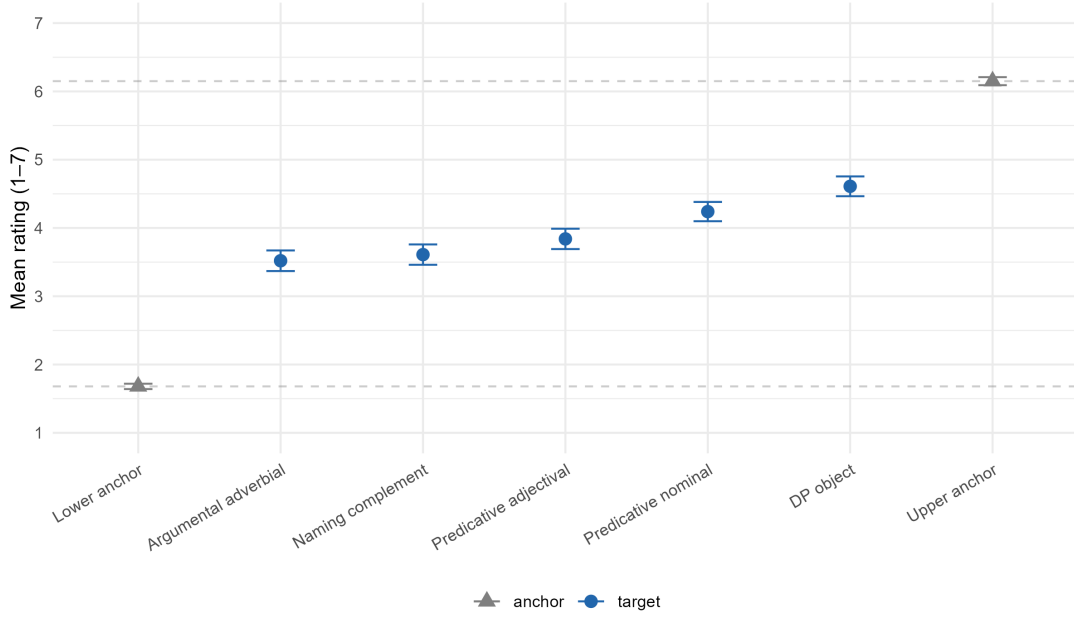
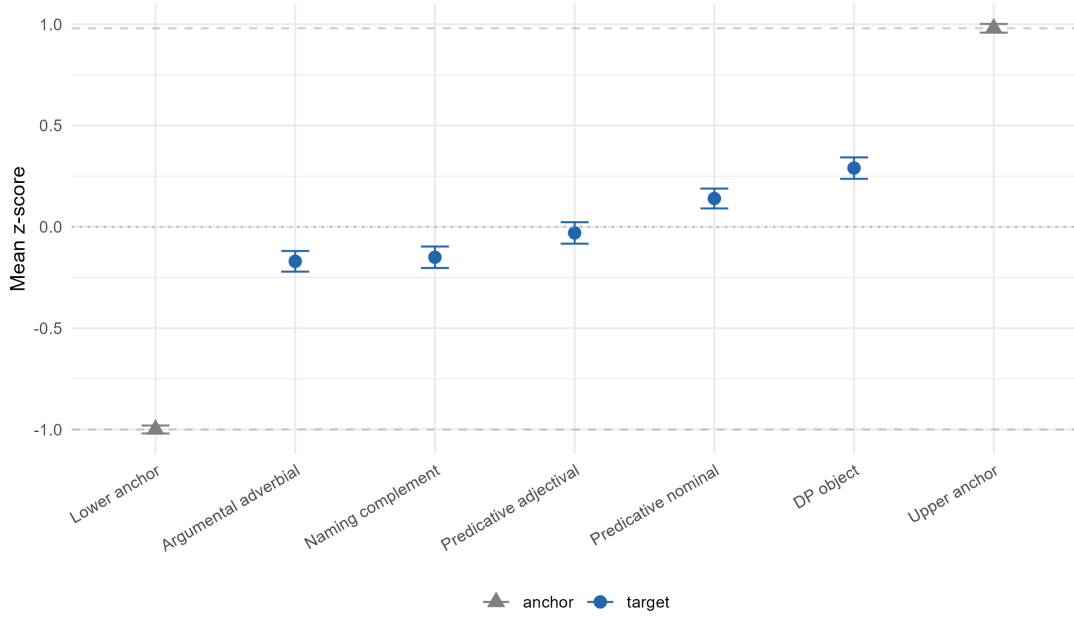


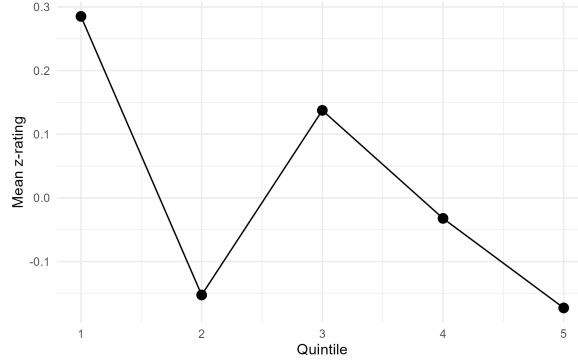
Figure 2: Mean acceptability ratings ($\pm 95\%$ CI) by condition (z-normalized).



three conditions clustered at lower values: PREDICATIVE ADJECTIVAL (3.84, $\bar{z} = -0.03$), NAMING COMPLEMENT (3.61, $\bar{z} = -0.15$), and ARGUMENTAL ADVERBIAL (3.52, $\bar{z} = -0.17$).

A significant negative correlation between trial order and z-score-normalized ratings was observed (Spearman’s $\rho = -0.193$, $p < .001$), Figure 3, indicating a small but reliable fatigue/calibration effect over the course of the experiment. However, this covariate was not included in the model due to insufficient identification with 8 items per condition in a between-items design. The fatigue/calibration effect should be noted as a limitation, but not a threat to the validity of the main contrasts as it is presumably distributed evenly across all conditions and the trial order $\times$ condition correlation did not show significance.

Figure 3: Order effect diagnostic (z-normalized).



3.3 Mixed-effects model

Ratings were z-score normalized per participant across all trials prior to analysis. A linear mixed-effects model was fitted to the z-scored target ratings using the `lme4` package (Bates et al., 2015) with Satterthwaite’s approximation for degrees of freedom (Kuznetsova et al., 2017). The fixed effect structure included CONDITION (5 levels; treatment-coded with DP OBJECT as reference). The random effect structure followed a maximal-to-minimal pre-registered fallback procedure (Barr et al., 2013): models with full by-participant random slopes, uncorrelated slopes, slopes-only, and intercepts-only were fitted in sequence, with the most complex convergent non-singular model retained. All models with random slopes were singular; the intercepts-only model was therefore used:

$$\text{rating_z} \sim \text{condition} + (1 \mid \text{participant}) + (1 \mid \text{item_id}),$$

where `item_id` encodes unique stimulus identity (40 levels; 5 conditions $\times$ 8 items). Model fit: REML criterion = 4606.5; random effect standard deviations: participant intercept = 0.207, item intercept = 0.165, residual = 0.553.

3.4 Confirmatory contrasts

Estimated marginal means were obtained via the `emmeans` package (Lenth & Piaskowski, 2026). Four pre-registered directional contrasts (*H1a–H1d*: DP OBJECT vs. each non-*e*-type condition) were tested with Holm–Bonferroni correction controlling the family-wise Type I error rate at $\alpha = .05$. Effect sizes were computed as Cohen’s $d = \hat{\delta}/\hat{\sigma}$, where $\hat{\sigma}$ is the model residual standard deviation. Results are reported in Table 2.

H1a, *H1c*, and *H1d* were confirmed: DP OBJECT was rated significantly more acceptable than NAMING COMPLEMENT, PREDICATIVE ADJECTIVAL, and ARGUMENTAL ADVERBIAL, with medium-to-large effect sizes ($d = 0.57$ – 0.83). *H1b* was not confirmed: the contrast between DP OBJECT and PREDICATIVE NOMINAL did not reach significance ($d = 0.25$, $p = .133$). However, post-hoc power analysis indicates that the study was underpowered to detect effects of this magnitude, leaving open whether predicative nominals pattern with *e*-type or non-*e*-type arguments.

Table 2: Confirmatory contrasts: DP OBJECT vs. each non-*e*-type condition (Holm–Bonferroni corrected; family-wise $\alpha = .05$).

	Contrast	$\hat{\delta}$	<i>SE</i>	<i>df</i>	<i>t</i>	p_{Holm}	<i>d</i>
H1a	DP > Naming complement	0.443	0.091	30.9	4.87	< .001	0.80
H1b	DP > Predicative nominal	0.140	0.091	30.9	1.54	.133	0.25
H1c	DP > Predicative adjectival	0.313	0.091	30.9	3.44	.003	0.57
H1d	DP > Argumental adverbial	0.460	0.091	30.9	5.06	< .001	0.83

3.5 Exploratory contrasts among non-*e*-type conditions

As a pre-registered exploratory analysis, all pairwise contrasts among the four non-*e*-type conditions were computed with Holm–Bonferroni correction. Results are provided in Table 3.

Table 3: Exploratory pairwise contrasts among non-*e*-type conditions (Holm–Bonferroni corrected; family-wise $\alpha = .05$).

	Contrast	$\hat{\delta}$	<i>SE</i>	<i>df</i>	<i>t</i>	p_{Holm}
	Naming – Predicative nominal	−0.303	0.091	30.9	−3.33	.011
	Pred. nominal – Argumental	0.320	0.091	30.9	3.51	.008
	Naming – Predicative adjectival	−0.131	0.091	30.9	−1.43	.348
	Pred. nominal – Pred. adjectival	0.173	0.091	30.9	1.90	.269
	Pred. adjectival – Argumental	0.147	0.091	30.9	1.62	.348
	Naming – Argumental	0.017	0.091	30.9	0.18	.856

Two significant contrasts emerged: PREDICATIVE NOMINAL was rated higher than both NAMING COMPLEMENT ($p = .011$) and ARGUMENTAL ADVERBIAL ($p = .008$). No other pairwise differences among non-*e*-type conditions reached significance. This suggests a two-cluster structure within the non-*e*-type conditions, with PREDICATIVE NOMINAL forming a distinct intermediate group, while NAMING COMPLEMENT, PREDICATIVE ADJECTIVAL, and ARGUMENTAL ADVERBIAL are mutually indistinguishable.

4 Discussion

The results replicate and extend the core finding of Kasenov & Rudnev (2024): the semantic type of the elided argument significantly affects the acceptability of Russian verb-stranding constructions. Three of four *e*-type vs non-*e*-type contrasts (*H1a*, *H1c*, *H1d*) were confirmed with medium-to-large effect sizes, consistent with AE as the primary analysis for verb-stranding constructions in Russian.

The present study, however, goes beyond a binary *e*-type vs. non-*e*-type opposition by decomposing the non-*e*-type category into four distinct subtypes. The results reveal that this category is internally heterogeneous. PREDICATIVE NOMINAL complements (e.g., *stat’ učitelem* ‘become a teacher’, *sčitat’ favoritom* ‘consider a favourite’) pattern with DP OBJECT rather than with the remaining non-*e*-type conditions: *H1b* was not confirmed ($d = 0.25$, $p = .133$), and PREDICATIVE NOMINAL was rated significantly higher than NAMING COMPLEMENT and ARGUMENTAL ADVERBIAL in the exploratory analysis. By contrast, NAMING COMPLEMENT, PREDICATIVE ADJECTIVAL, and ARGUMENTAL ADVERBIAL form a uniform lower-acceptability cluster, mutually indistinguishable in pairwise comparisons.

Before interpreting the non-significant contrast, a methodological remark is necessary. Post-hoc power analysis based on the observed variance components indicates that the study had approximately 22% power to detect an effect of the observed size ($d = 0.25$) at the Holm-corrected threshold, and approximately 44% power at uncorrected threshold. The minimum detectable effect size at 80% power was $d \approx 0.53$. Thus the non-significant result for *H1b* cannot be taken as evidence that predicative nominal patterns with DP OBJECT; it may simply reflect insufficient sensitivity. The theoretical discussion below should be read with this limitation in mind.

That said, if better-powered studies confirm the elevated acceptability of PREDICATIVE NOMINAL,

one possible account is that Russian predicative nominals are ambiguous between a genuine small-clause predicate reading (type $\langle e, t \rangle$) and an individual-denoting reading (type e) in which the nominal refers to the kind or role rather than to a property (cf. Landau, 2020). Under the latter reading, the elided argument would qualify as e -type, rendering the verb-stranding construction grammatical. This analysis would predict gradient acceptability rather than a categorical distinction, which is consistent with the intermediate z-scores observed for PREDICATIVE NOMINAL ($\bar{z} = 0.14$, between DP at 0.29 and the lower cluster at -0.03 to -0.17). Alternatively, the predicative nominal items in this experiment involved high-frequency collocations, which may be processed more holistically than compositionally. If participants treat such collocations as semi-fixed units, verb-stranding may be facilitated not because of the true argument type, but because the entire VP is retrieved from the antecedent conjunct as a single chunk. Disentangling the semantic-type account from a frequency-based account would require future experiments that systematically vary the collocational strength of predicative nominal items.

The convergent low acceptability of NAMING COMPLEMENT, PREDICATIVE ADJECTIVAL, and ARGUMENTAL ADVERBIAL is consistent with Landau’s generalization that non- e -type arguments block argument ellipsis cross-linguistically as they do not denote individuals: naming verbs (e.g., *nazvat’* ‘name’, *prozvat’* ‘nickname’) introduce a propositional small clause, predicative adjectivals denote properties (type $\langle e, t \rangle$), and argumental adverbials (e.g., *vesti sebja xorošo* ‘behave well’) involve a manner argument. The uniform degradation across these three types supports the view that the relevant constraint targets a broad semantic category (non- e -type) rather than any specific morphosyntactic configuration.

A potential confound should be acknowledged. In the NAMING COMPLEMENT and ARGUMENTAL ADVERBIAL conditions, the verb-stranding site involves two absent constituents (naming: the DP object and the naming predicate; argumental adverbial: the reflexive pronoun and the manner adverb), whereas in the predicative nominal and predicative adjectival conditions only a single constituent is absent. The two conditions with two missing elements are precisely the two lowest-rated conditions, raising the possibility that the degradation reflects not (only) semantic type but also the amount of unpronounced material. However, this confound does not undermine the core e -type diagnostic: DP OBJECT, which involves a single absent constituent, is rated significantly higher than PREDICATIVE ADJECTIVAL, which also involves a single absent constituent. Furthermore, if the number of missing constituents were the sole driver, PREDICATIVE ADJECTIVAL would be expected to pattern with PREDICATIVE NOMINAL rather than with NAMING COMPLEMENT and ARGUMENTAL ADVERBIAL; this is not supported by the observed data (PREDICATIVE ADJECTIVAL vs. NAMING COMPLEMENT: $p = .348$; PREDICATIVE ADJECTIVAL vs. ARGUMENTAL ADVERBIAL: $p = .348$). Nonetheless, fully disentangling the effects of semantic type and ellipsis size would require a design that crosses these factors independently.

A further potential limitation concerns polarity ellipsis (TP-ellipsis in the sense of Gribanova (2017)) as a competing parse. The additive context established by *tože* makes this parse pragmatically unlikely. And as Kasenov & Rudnev (2024) report a comparable e -type effect in both matrix and embedded environments, we suggest that the semantic type constraint holds independently of polarity ellipsis availability. Even if such a parse were marginally available, it would apply equally across all five conditions and could not account for the observed contrasts.

The absence of a significant difference between NAMING COMPLEMENT and ARGUMENTAL ADVERBIAL ($\hat{\delta} = 0.017$, $p = .856$) is also noteworthy, given that these two conditions differ substantially in their surface morphosyntactic properties: naming complements are nominal, while argumental adverbials are adverbial. The convergence of their acceptability profiles supports the idea that the relevant property is semantic type rather than morphosyntactic category. At the same time, as noted above, both conditions also share the property of involving two absent constituents, and the convergence may partly reflect this structural parallel.

5 Conclusion

This study tested the semantic type constraint on verb-stranding constructions in Russian by comparing acceptability ratings for an e -type argument (DP OBJECT) with four non- e -type argument types. Three of four pre-registered contrasts were confirmed: DP OBJECT was rated significantly higher than NAMING COMPLEMENT, PREDICATIVE ADJECTIVAL, and ARGUMENTAL ADVERBIAL, corroborating the AE analysis for Russian verb-stranding, under the assumption that Landau’s e -type generalisation holds. The non-significant contrast with PREDICATIVE NOMINAL, together with its intermediate ac-

ceptability profile, points to a potential *e*-type construal of predicative nominals that warrants further investigation.

Beyond the binary *e* vs. non-*e* distinction, the experiment reveals that the non-*e*-type category is not monolithic: PREDICATIVE NOMINAL patterns apart from the remaining three conditions, which form a lower-acceptability cluster. This internal heterogeneity suggests that future work on AE should attend to finer-grained semantic distinctions within the non-*e*-type domain.

Several limitations should be noted. The design used 8 items per condition, leading to a limited amount of item-level degrees of freedom; post-hoc power analysis indicates that the study was underpowered to detect small effects. The availability of TP-ellipsis as a competing parse was not controlled. A trial order effect was observed, though it is unlikely to have differentially affected the conditions. Future experiments with a larger item set, a control for TP-ellipsis, and a wider range of predicative nominals would strengthen the universality of the present findings.

Note: the analysis plan for this experiment was pre-registered: <https://osf.io/nt6gq/> (registration DOI: 10.17605/OSF.IO/NT6GQ). Following the principles of open science, the full R (R Core Team, 2026) analysis script and the full stimulus list are provided on the OSF project page associated with this study: <https://osf.io/nkg8y>.

References

- Bailyn, John Frederick. 2017. Against a VP ellipsis account of Russian verb-stranding constructions. In McClure, Alexander, William T. Vovin (ed.), *Studies in Japanese and Korean historical and theoretical linguistics and beyond: Festschrift presented to John B. Whitman*, 93–109. Brill.
- Bailyn, John Frederick & Bondarenko, Tatiana. 2021. Russian. In van Craenenbroeck, Tanja, Jereon Temmerman (ed.), *The Oxford handbook of ellipsis*, Oxford: Oxford University Press.
- Barr, Dale J. & Levy, Roger & Scheepers, Christoph & Tily, Harry J. 2013. Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language* 68(3). 255–278.
- Bates, Douglas & Mächler, Martin & Bolker, Ben & Walker, Steve. 2015. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67(1). 1–48.
- Bošković, Željko. 2018. On pronouns, clitic doubling, and argument ellipsis: Argument ellipsis as predicate ellipsis. *English Linguistics* 35(1). 1–37.
- Goldberg, Lotus Madelyn. 2005. *Verb-stranding vp ellipsis: A cross-linguistic study*: McGill University dissertation.
- Gribanova, Vera. 2013a. A new argument for verb-stranding verb phrase ellipsis. *Linguistic Inquiry* 44(1). 145–147.
- Gribanova, Vera. 2013b. Verb-stranding verb phrase ellipsis and the structure of the Russian verbal complex. *Natural Language & Linguistic Theory* 31(1). 91–136.
- Gribanova, Vera. 2017. Head movement and ellipsis in the expression of Russian polarity focus. *Natural Language & Linguistic Theory* (35). 1079–1121.
- Kasenov, Daniyar & Rudnev, Pavel. 2024. Experimental evidence against verb-stranding VP ellipsis in Russian. Ms., HSE University.
- Kuznetsova, Alexandra & Brockhoff, Per B. & Christensen, Rune H. B. 2017. lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software* 82(13). 1–26.
- Landau, Idan. 2018. Missing objects in Hebrew: Argument ellipsis, not VP ellipsis. *Glossa: A Journal of General Linguistics* 3(1).
- Landau, Idan. 2020. On the nonexistence of verb-stranding vp-ellipsis. *Linguistic Inquiry* 51(2). 341–365.

- Landau, Idan. 2023a. Argument ellipsis as external merge after transfer. *Natural Language & Linguistic Theory* 41(2). 793–845.
- Landau, Idan. 2023b. Type-restricted argument ellipsis and generalized quantifiers. *Linguistic Inquiry* 56(1). 97–129.
- Lenth, Russell V. & Piaskowski, Julia. 2026. *emmeans: Estimated marginal means, aka least-squares means*. <https://rvlenth.github.io/emmeans/>.
- Oku, Satoshi. 1998. *A theory of selection and interpretation of null nominal categories*: University of Connecticut dissertation.
- R Core Team. 2026. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing Vienna, Austria. <https://www.R-project.org/>.
- Saito, Mamoru. 2007. Notes on East Asian argument ellipsis. *Language Research* 43(2). 203–227.
- Takahashi, Daiko. 2006. Apparent parasitic gaps and null arguments in Japanese. *Journal of East Asian Linguistics* 15. 1–35.
- Takahashi, Daiko. 2008. Quantificational null objects and argument ellipsis. *Linguistic Inquiry* 39(2). 307–326.
- Zehr, Jeremy & Schwarz, Florian. 2018. PennController for Internet based experiments (IBEX). <https://doc.pciibex.net>.